Introduction to Composites Material

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Polymer Matrix composites

Functions of the Matrix Material (Primary Phase)

- It takes the load and transfer it to the reinforcement

- It binds or holds the reinforcement phase and protects the same from external damage (either chemical and mechanical)

- Matrix also separates the individual fibers and prevents brittle cracks from passing completely across the section of the composite

Polymers are widely used matrix materials for industrial and advanced composites because of

- ✓ low density
- ✓ Ease of fabrication
- ✓ Reasonably good mechanical and electrical properties
- ✓ Good corrosion resistance
- ✓ Adequate flexibility

Polymer matrix composites (PMC) is the material consisting of a polymer (resin, as primary phase) matrix combined with a fibrous reinforcing dispersed phase (secondary phase)

or

Polymer matrix composites (PMC) are materials made up of fibers that are embedded in an organic polymer matrix. These fibers are introduced to enhance selected properties of the material

Polymer matrix composites are classified based on their level of strength and stiffness into two distinct types:

1. Reinforced plastics - confers additional strength by adding embedded fibrous matter into plastics

2. Advanced Composites - consists of fiber and matrix combinations that facilitate strength and superior stiffness. They mostly contain high-performance continuous fibers such as high-stiffness glass (S-glass), graphite, aramid or other organic fibers

Properties of some high-performance polymer matrices of composites

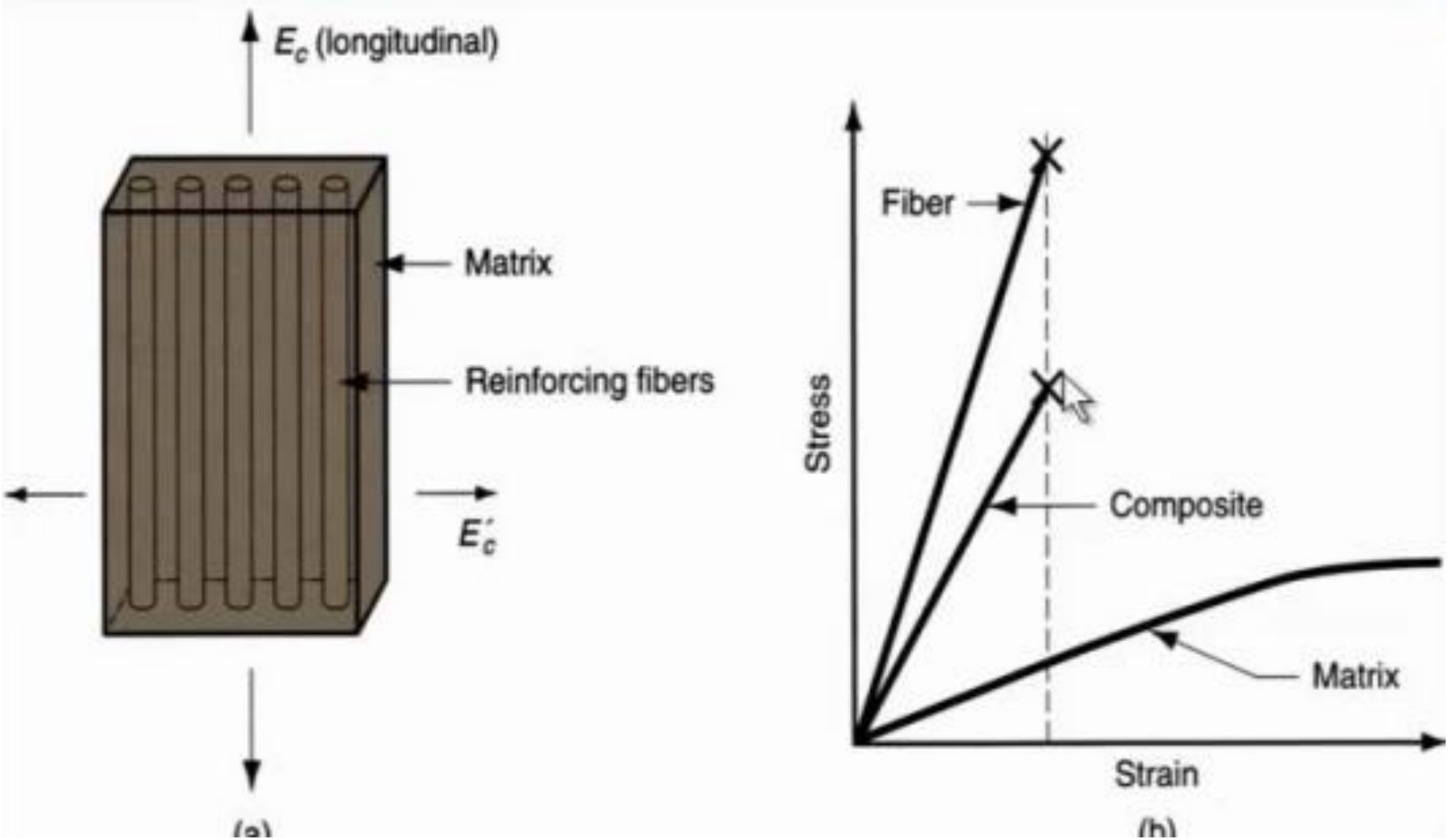
Properties	*Epoxy*	*Polyimide (Thermoset)*	*Polyimide (Thermoplastic)*	*Poly-sulphone*	*PEEK*
Density, kg m^{-3}	1110–1400	1430	—	1240	1300–1320
Modulus of elasticity, GPa	2.41–4.2	3.2	2.07–2.76	2.5	3.9
Tensile strength, MPa	27.6–90	56	—	70	70.3–103.5
Compressive strength, MPa	140	187	—	96	—
CTE, $10^{-6}(°C)^{-1}$	81–117	27–90	81–100	—	72–85
Thermal conductivity, $W(mK)^{-1}$	0.19	0.23–0.50	0.096–0.18	—	0.25
T_g, °C	130–250	370	280–330	185	143

Properties of PMCs

The constituents of a PMC, which affect its overall properties, are

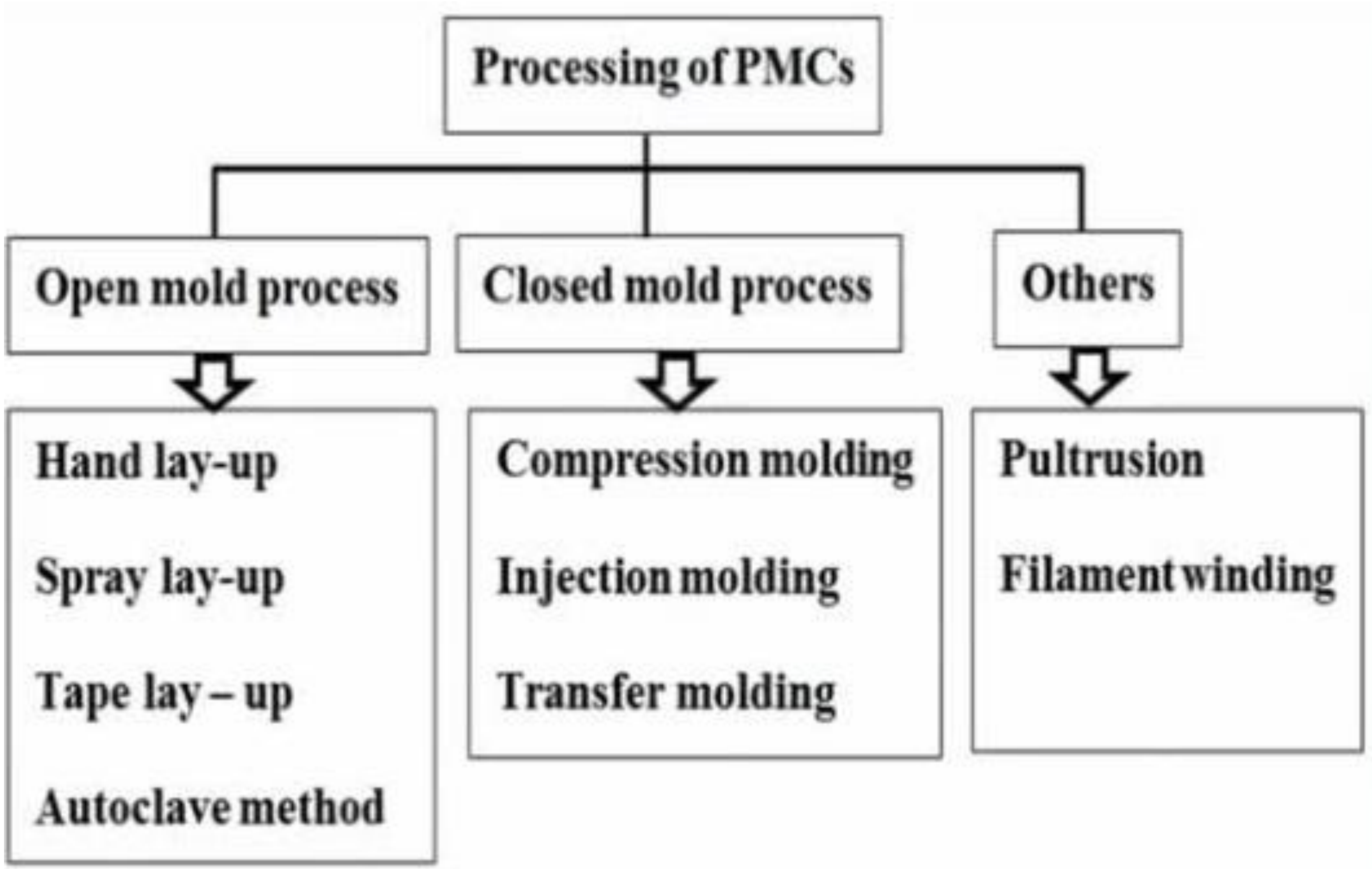
- **Matrix** - This is the polymer, which is a continuous phase and is classified as the weak link in a PMC structure.
- **Reinforcement** - This is a discontinuous phase and is a principal load-bearing component. It can either be glass, quartz, basalt, or carbon fiber.
- **Interphase** - The interphase between the reinforcement and matrix phases where load transmission takes place.

Aside from the types of matrix and reinforcement used, other factors affecting the properties of a PMC are the constituents' relative proportions, the reinforcement geometry and the nature of the interphase



E_c (longitudinal)
Matrix
Reinforcing fibers
E_c'
Fiber
Stress
Composite
Matrix
Strain

Processing of polymer matrix composites



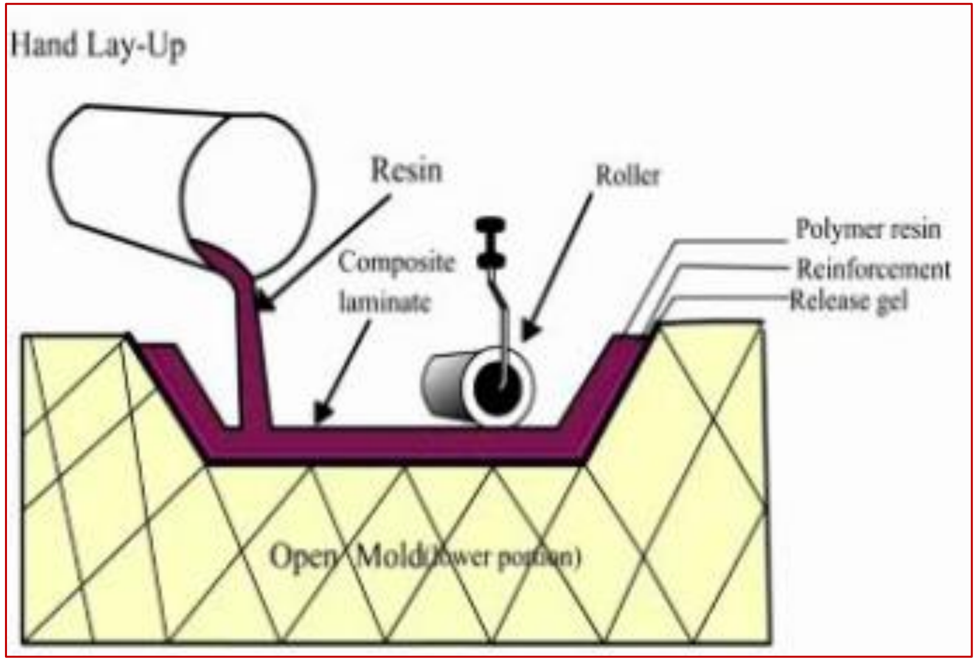
Hand layup technique

- ❑ It is the simplest method of composite processing.
- ❑ The infrastructural requirement or this method is also minimal
- ❑ The processing steps are quite simple and easy to understand.
- ❑ Production rate is less

Raw materials

Matrix: Epoxy, Polyester, polyvinyl ester, phenolic resin, polyurethane resin

Reinforcement: Glass fiber, carbon fiber, Kevlar, natural plant fibers.



Limitations:

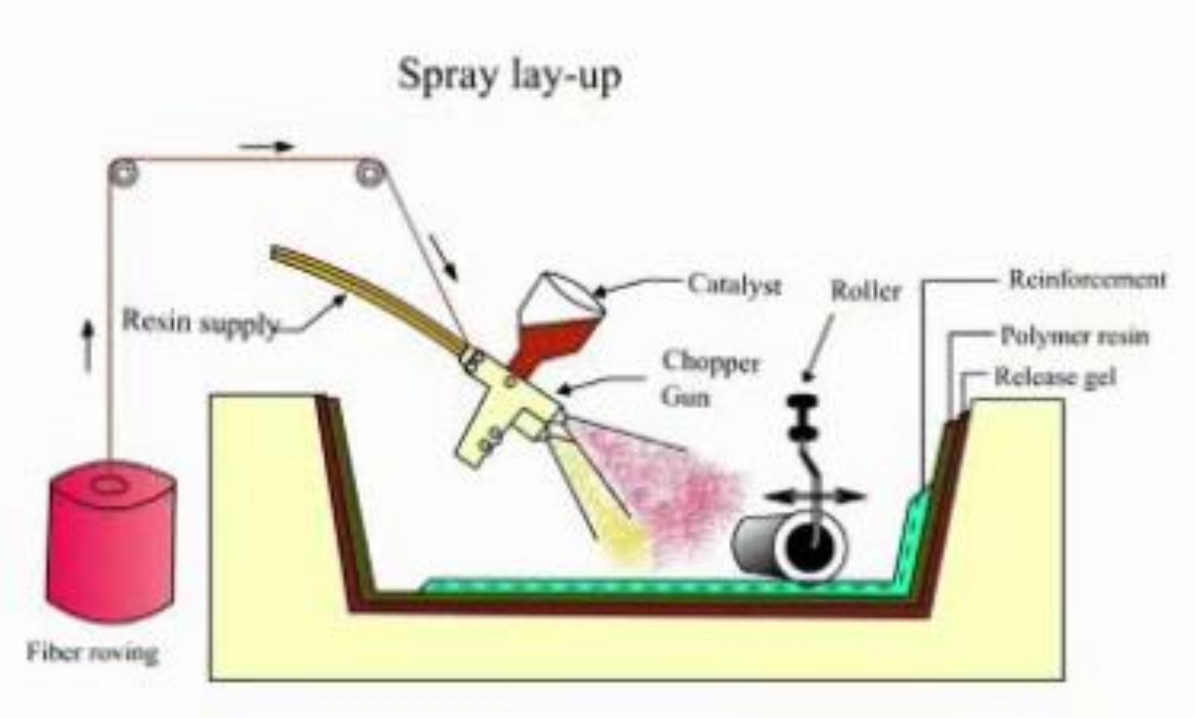
- Mainly suitable for thermosetting polymers
- The resin must have low viscosity
- Health and safety consideration of resins.
- Applications.

Applications:

- Aircraft components
- Automotive parts
- Boat hulls
- Dash board

Spray layup

- It is an open mold process
- An extension of the hand lay up method
- Processing is similar to hand lay up process, the only difference lies in application of resin and fiber to the mold
- The process is well suitable for small to medium volume production
- The process is cost effective for producing small and large parts.



ADV:

- Provides high volume fraction of reinforcement in composites
- Processing is fast

  Tolling cost is low
- It achieves better wetting of the fiber with fewer voids than with hand lay up
- There is no part size limitation in this technique

Limitation

- Poor roll out can include structural weakness by leaving air bubbles, dislocation of fibers and poor wet out
- It is not suitable for parts having high structural requirement.
- Fiber volume fraction is difficult to control
- Th eprocess is highly dependent on operator skill.

Resin Transfer molding

- It is a closed molding process
- Also known as liquid transfer molding process
- Resin is transferred over the already placed reinforcement
- The process is effective for production of structural parts with low cost; in low to medium production quantities.
- It is possible to achieve near net shape with controlled fiber orientation. Generally continuous fibers are used in this method.

Raw materials:

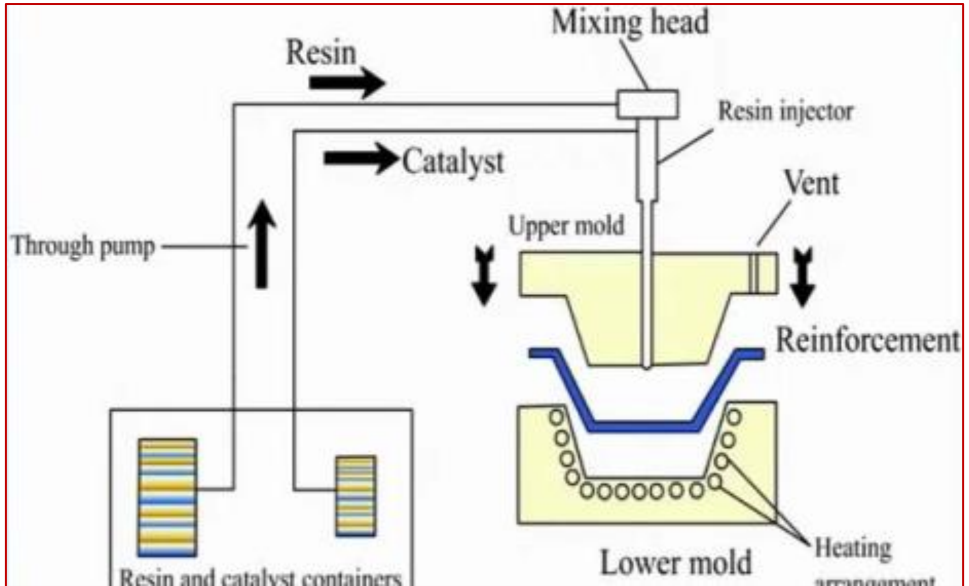
Reinforcing materials: Glass fiber, carbon fiber, aramid fiber, natural plant fibers.

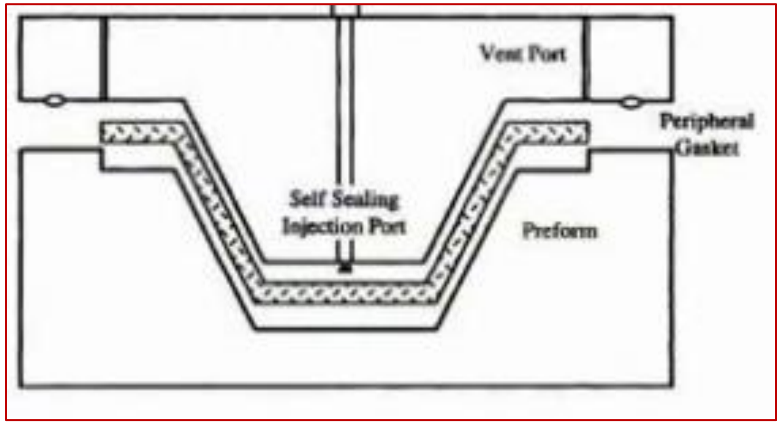
Matrix: Epoxy, methyl methacrylate, Polyester, Polyvinyl ester, phenolic resin

Resin Transfer molding Procedure

- A release gel is applied on the mold surface for easy removal of the composite.
- Reinforcement is in terms of either woven mat or chopped fiber mat form is placed on the surface of lower half mold.
- The mold is properly closed and clamped.

  The resin is pumped into the mold through ports and air is displaced through other vents.
- The uniformity of resin flow can be enhanced by using a catalyst as a accelerator and vacuum application.
- After curing, the mold is opened and composite product is taken out.





Controlling parameters

- Resin viscosity
- Injection time
- Injection pressure
- Temperature of the mold

Applications

Complex structures can be produced

Automotive body parts, big containers, bathtubs

ADV:

- Composite parts produced with this method has good surface finish on both the surfaces of the product
- Any combination of reinforced materials in any orientation can be achieved.
- Process can be manual control, semi-automated or highly automated.
- Composite part thickness is uniform
- Strict dimensional tolerances are possible to achieve
- Ability to incorporate inserts and other attachments into molding.
- Higher production rate is possible with automation.


THANK YOU